



- 4.1 Basic of forces and
free body diagram (FBD)
- 4.2 Newton's Laws of Motion



4.1 Basic of forces and free body diagram

At the end of this chapter, students should be able to:

- (a) **Identify** the forces acting on a body in different situations:
 - Weight
 - Tension
 - Normal force
 - Friction
 - External force (pull or push)
- (b) **Sketch** free body diagram.
- (c) **Determine** static friction and kinetic friction.

$$f_s \leq \mu_s N, f_k = \mu_k N$$

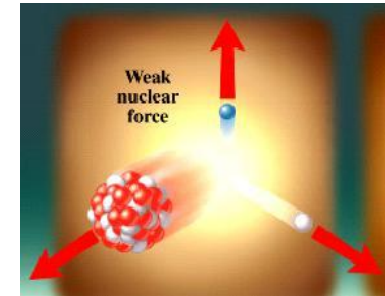
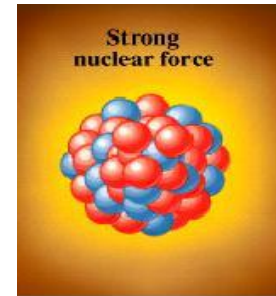
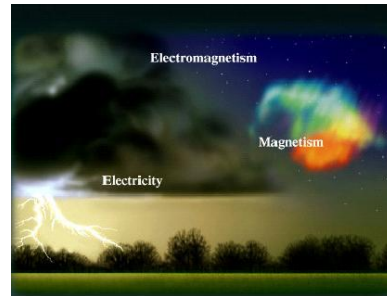
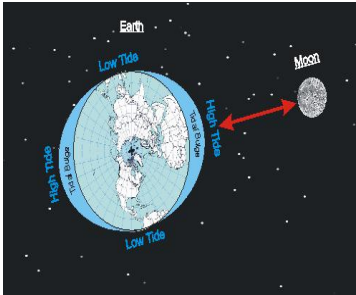


4.2 Newton's Laws of Motion

At the end of this chapter, students should be able to:

- a) **State** Newton's law of motion.
- b) **Apply** Newton's law of motion.

(Experiment 5 : Friction)



Gravitational forces

- The forces involve attraction between massive body.
 - is a long-range forces.
 - the weakest forces in nature.

Electromagnetic forces

- The attractive and repulsive forces between electric charges.
- is a long-range forces.

Strong nuclear forces

- The attractive forces bonding neutron and proton in atomic nucleus.
- is a short-range forces and the strongest forces in nature

Weak nuclear forces

- cause the unstable condition for atomic nucleus and is responsible for the radioactive decay.
- is a short-range forces and 12 times weak compare with electromagnetic forces.

What is FORCE?

- A force is **a push or pull** on the object
- It change the direction of all objects
- A force is a **vector** quantity. It has both a **magnitude** and a **direction**
- SI unit of force, $\vec{F}$ is kg m s^{-2} or **Newton (N)**
- Anytime you see something moving, you can be sure that a force created its motion
- Forces are everywhere
- Without forces activities like sports, dancing, driving..basically everything would be imposibble



TYPE OF FORCES

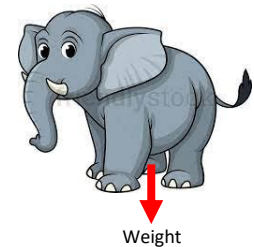
EXTERNAL FORCE

Object is be push or be pull is called external force



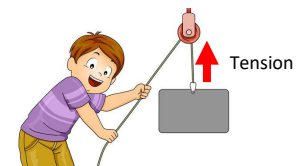
WEIGHT

Every object will experience weight because they have mass



TENSION

Every tied object by a rope or cord will experience tension



FRICTION

Force that resist motion between two surfaces and it happen on a rough surface



NORMAL

Any object in touch with the surface will experience normal



WEIGHT

- Force exerted on a body due to **gravitational pull**.

$$W = mg$$

weight ← ← gravity acceleration (9.81 ms^{-2})
← mass (kg)

Unit: Newton (**N**)

Direction is always **vertically downward**

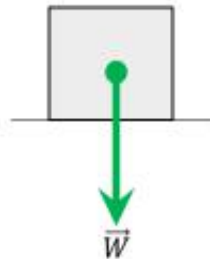
- Weight also known as **gravitational force**



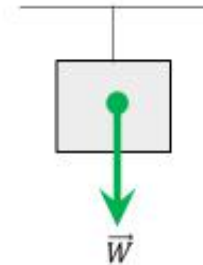
Direction of weight based on different situations



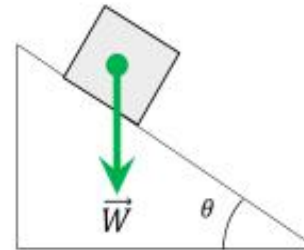
Situation 1: Cat on the chair



Situation 2: Hanging Pendant Lamp



Situation 3: Girl on the slide



TENSION (T)

Contact force exerted by a cord as it pulls on an object.

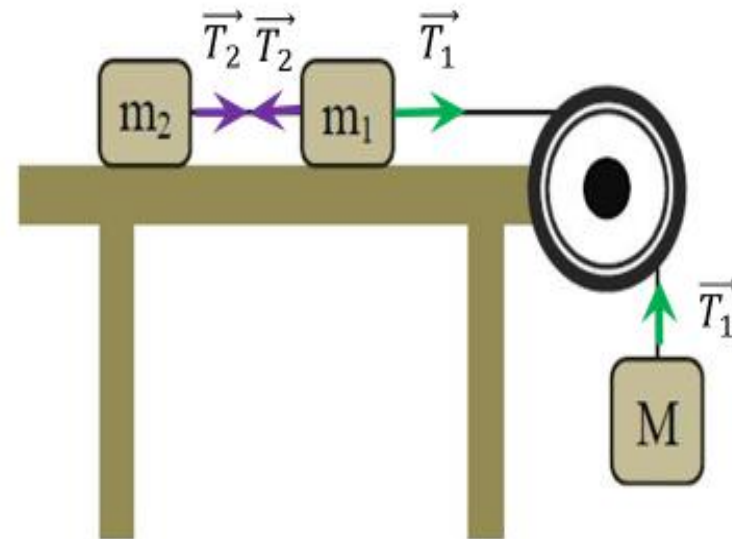
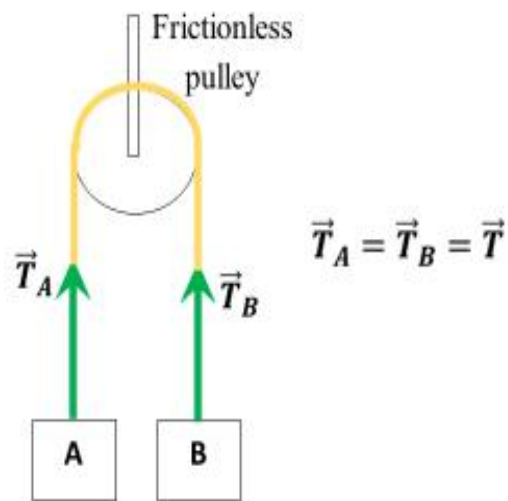
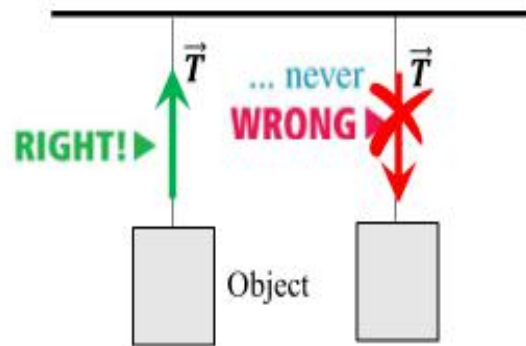
It is always directed away from the object and along the cord.

Unit: Newton (N)

- The cord has negligible mass, unbreakable, un-stretchable, and frictionless
- The tension is the same at all points in the rope. (**Same string possessed same tension.**)
- e.g. of cord: string, chain, wire, cable or etc.



Direction of tension



Direction of tension force is **always away from the object.**

SAME STRING/CORD has the **SAME TENSION**

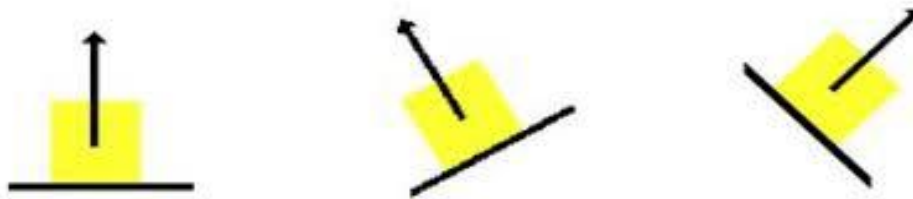
NORMAL

Normal force is defined as a **reaction force** that is **exerted by the surface** to an **object in contact with the surface**

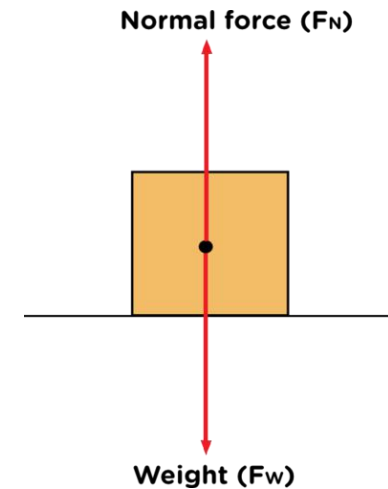
Always **directed perpendicular** to the surface

Unit: Newton (N)

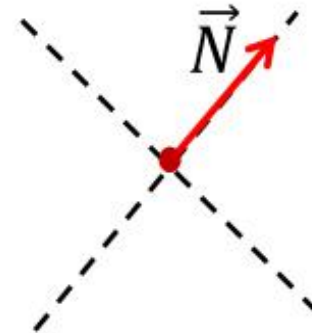
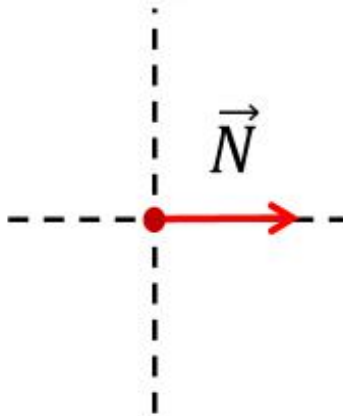
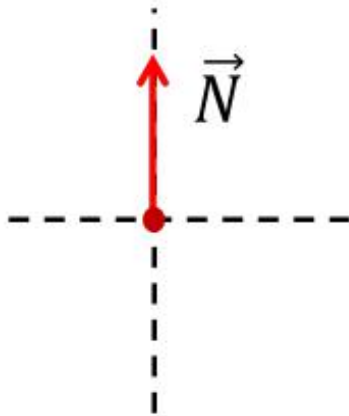
Normal force, **N** **NOT** always has to be same as weight, **W**



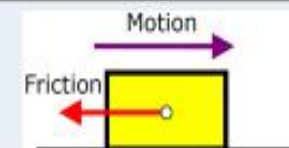
Normal forces are always directed perpendicular to the surface.



Direction of normal



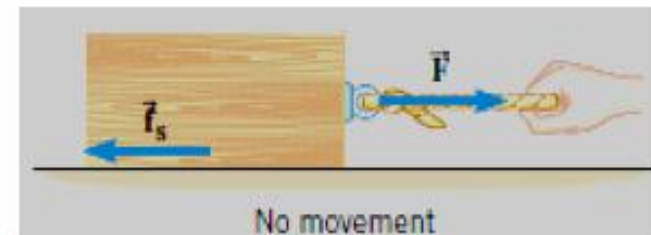
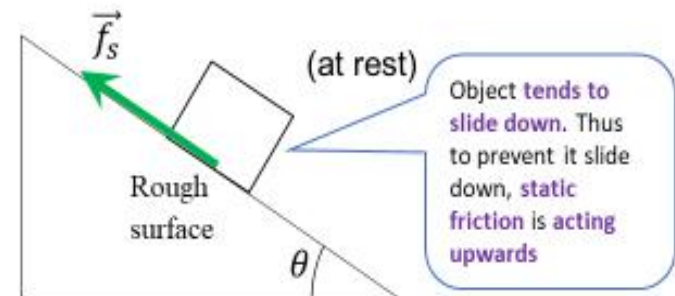
FRICITION



Definition:	Friction is defined as a force that opposes the relative motion of two surfaces in contact.
Direction:	acts PARALLEL to the surface in contact and in the OPPOSITE direction of the motion (or tendency of motion)
Unit:	Newton (N)
Remarks:	○ Friction is directly proportional to the reaction force. $f \propto N$ ○ Equation : $f = \mu N$ where f: friction force ; μ : coefficient of friction ; N : normal force ○ Coefficient of friction, μ is defined as the ratio between frictional force to normal force. $\mu = \frac{f}{N}$ ○ μ depends on the nature of the surfaces. ○ There are two types of friction: (i) Static friction, $\vec{f}_s$ (ii) Kinetic friction, $\vec{f}_k$ ○ In general $\mu_s > \mu_k$ thus $\vec{f}_s > \vec{f}_k$

STATIC FRICTION

Definition:	is the force that keeps an object “stuck” on a surface and prevents its motion. (frictional force before the object start moving, $v = 0$)
Direction:	Static friction is parallel to surface in contact and points opposite the direction of tendency of motion .
Remarks:	May have different values up to some maximum. $\vec{f}_s \leq \mu_s N$ Just before an object starts to slide (verge of motion), $\vec{f}_s$ is maximum $f_{s(max)} = \mu_s N$ μ_s: coefficient of static friction N: Normal force



$$F < f_s$$

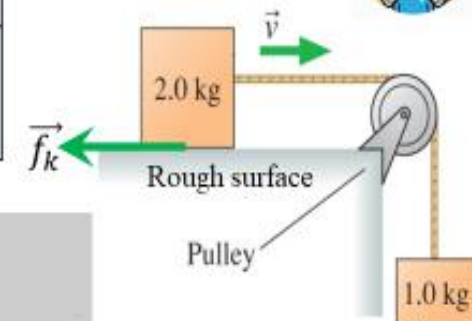
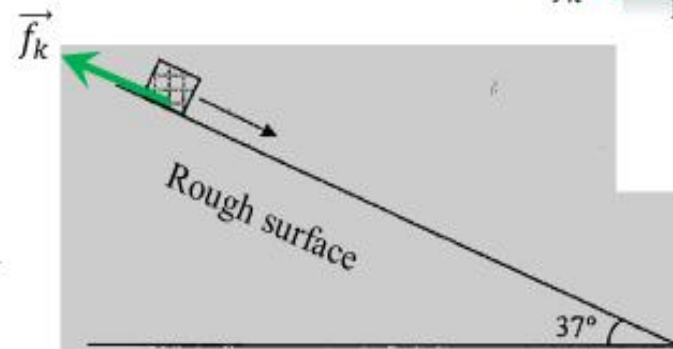
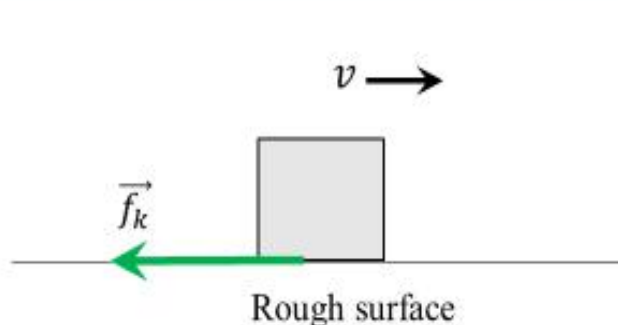


$$F = f_s$$

KINETIC FRICTION

Definition:	The frictional force exists between 2 object, when there is relative motion between at the interface of the surface contact.	
Formula:	$f_k = \mu_k N$	μ_k : coefficient of kinetic friction N : Normal force
Direction:	kinetic friction is parallel to surface in contact and points opposite direction of velocity ("the motion").	

- The agent for friction is **rough surface**.
- In our syllabus, the surface is always smooth **unless stated** rough surface or given value of μ .



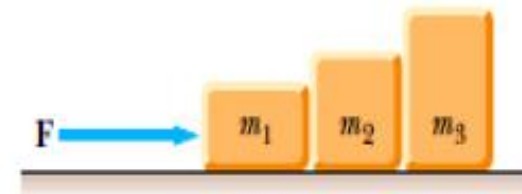
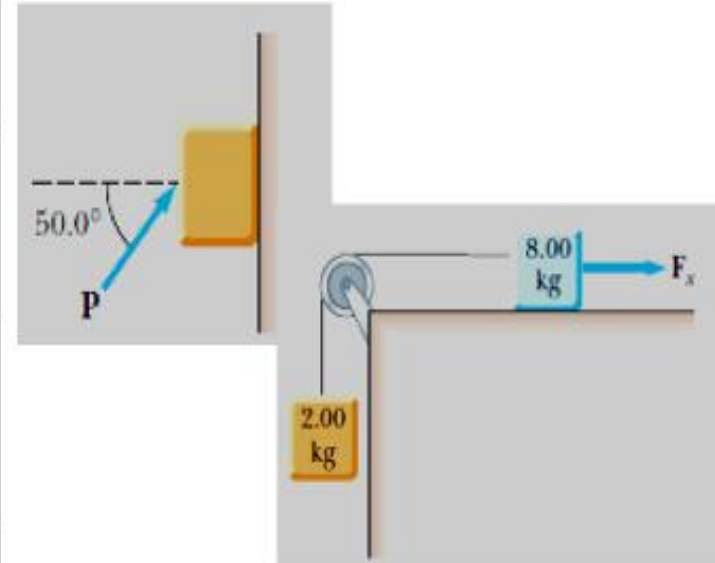
1.0 kg does not have any friction force because it **does not** contact with any rough surface

EXTERNAL FORCE

Definition	A force exerted on a system by an agency outside the system
Direction	Depending on the direction of application as stated in the question.
Unit	Newton (N)
Remarks	External force can either be push or pull force exerted on the object.

**CAUTION!**

Do NOT add in external force by your own.



Check Your Understanding

In physical science, a push and pull is called

- A) Motion
- B) Force
- C) Inertia

The force that causes resistance when rubbing sandpaper on wood is called

- A) Friction
- B) Weight
- C) Tension

If there is no frictional force, we cannot

- A) Speak
- B) Walk
- C) Eat

You throw a ball up in the air. As it travels upwards, gravity

- A) Causes the ball to speed up
- B) Causes the ball to slow down
- C) Does not affect the ball

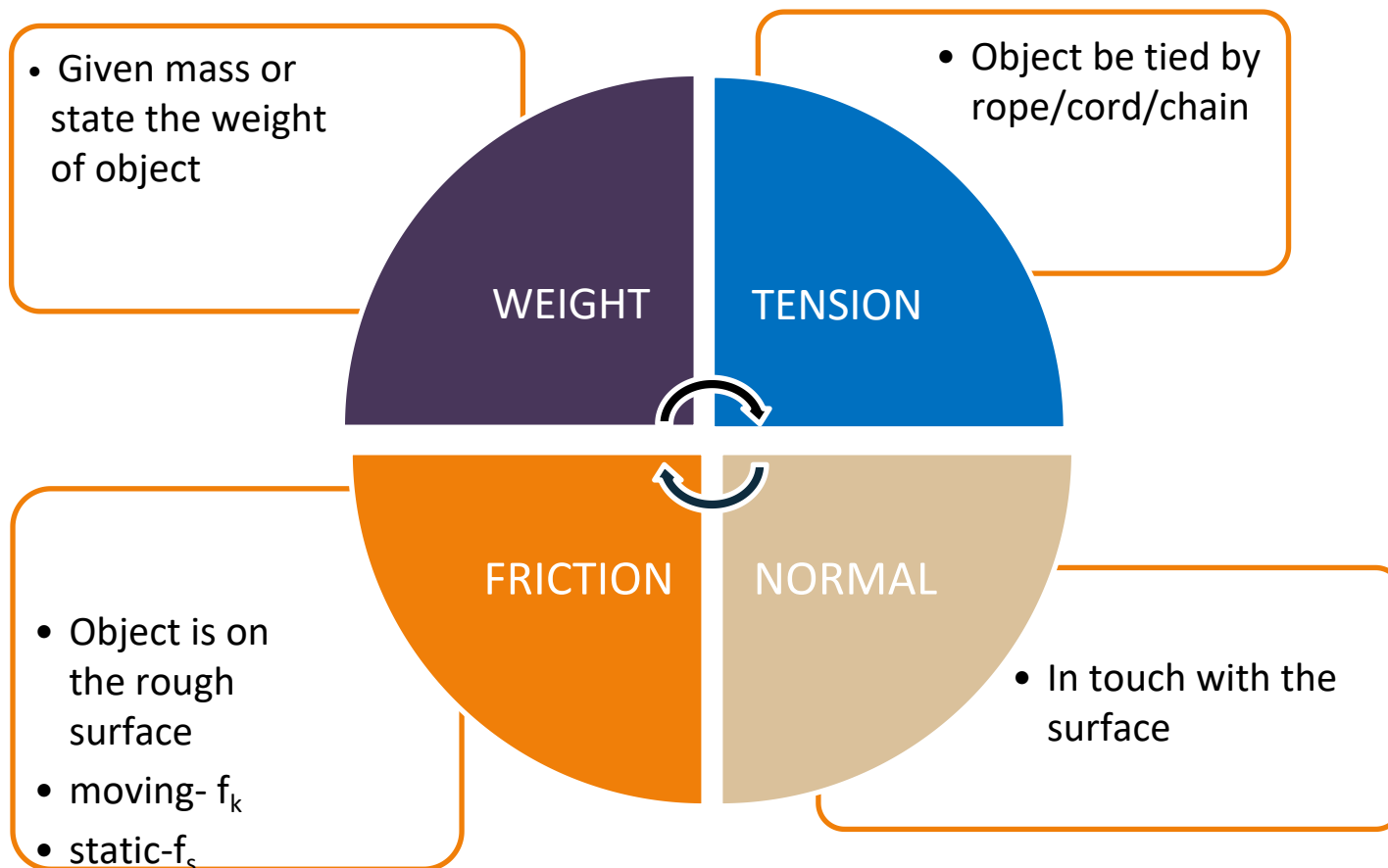
If a ball is rolling on the grass, it will slow down and stop because of

- A) Water resistance
- B) Speed
- C) Friction



Send your
answer to
your
lecturer

HOW TO IDENTIFY FORCES ACTING ON THE OBJECT?



For **external force**, just **redrawn as stated** in question !

Free body Diagram (FBD)

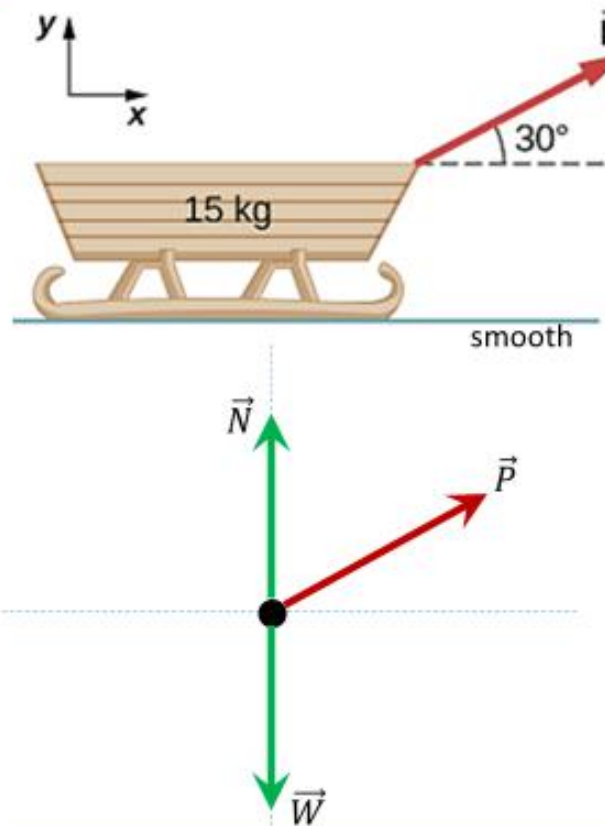
A diagram used to **show ALL forces acting** on upon an object in a given situation.

Drawing a free-body diagram

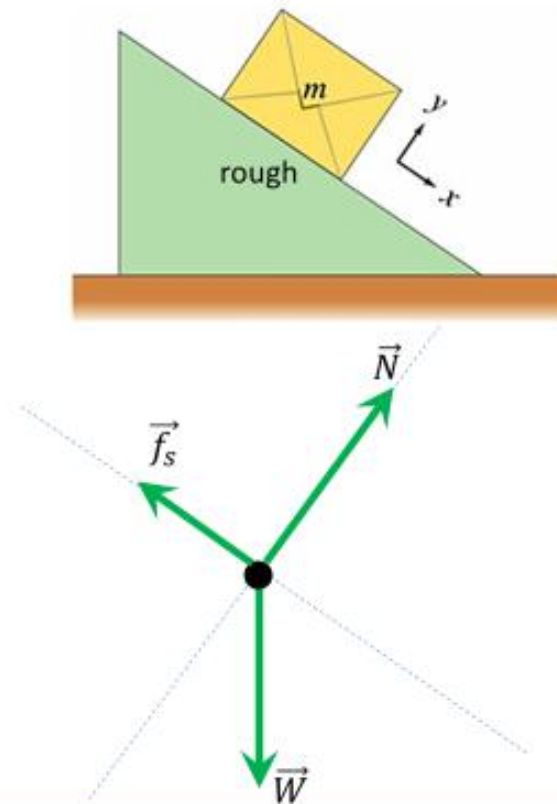
- 1) **Identify all forces** acting on the object.
- 2) **Draw a coordinate system.** Use the axes defined in your pictorial representation. If those axes are tilted, for motion along an incline, then the axes of the free-body diagram should be similarly tilted.
- 3) **Represent the object as a dot** at the origin of the coordinate axes.
- 4) **Draw** vectors representing each of the identified **forces**. Be sure to **label** each force vector. **CAUTION !** All **force vectors are drawn** from the point **outward** in the direction in which the forces are acting.



Case 1 : Horizontal surface

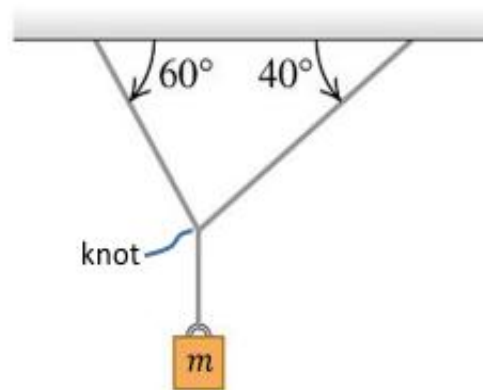


Case 2 : inclined plane

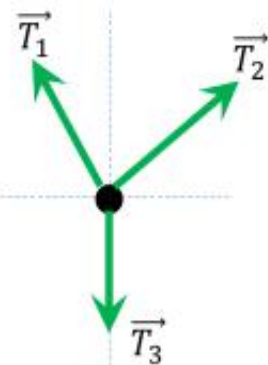
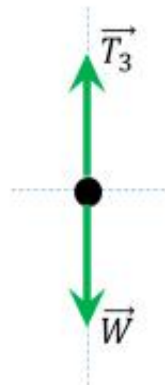




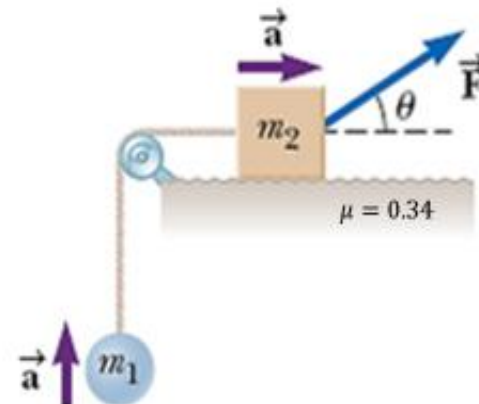
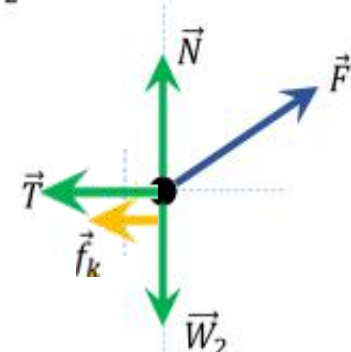
Case 3 : Hanging object



Knot:

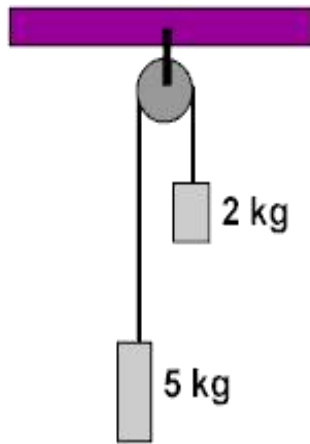
 m :

Case 4 : Pulley

 m_1 : m_2 :

Check Your Understanding

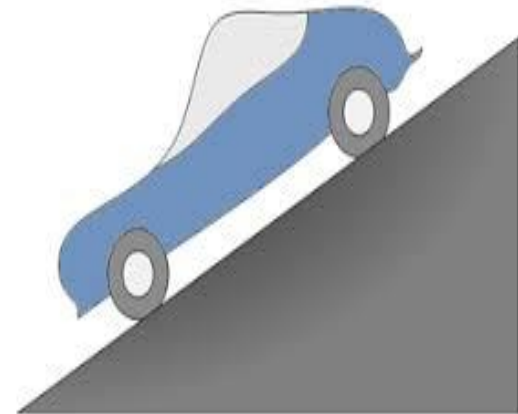
Draw Free Body Diagram based on the situation given



Case 1: Two object attach with a rope

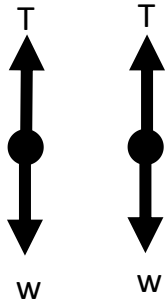
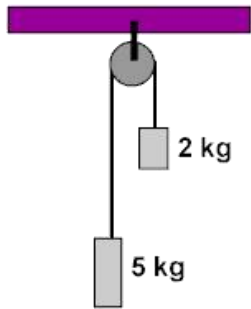


Case 2 : luggage being pull



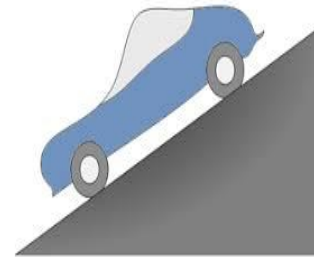
Case 3: Car park on slope hill

Answer:



object A

object B



4.2 Newton's laws of motion

Newton's first law

Net force also refer as resultant force, $\sum \vec{F}$



states "if there is **no net force acting** on an object (), object that is **at rest will remain at rest**, or an object that is **moving will continue to move** in straight line with **constant velocity**."

Newton's first law give the idea of inertia:

Inertia

- is defined as the **tendency of a body to resist change in motion**.
- depends on **mass**

Mass vs Weight

1 kg

1 kg object weigh only one-sixth as much as it did on earth, since gravity is weaker. But its mass will be the same.



NEWTON'S FIRST LAW

- a) A particle is said to be **in equilibrium** when the **vector sum of all forces acting on a particle (point) equals to zero**.
- c) Two types of equilibrium of a particle
- i. **Static** equilibrium ($v = 0$)
→ body at **rest (stationary)**.

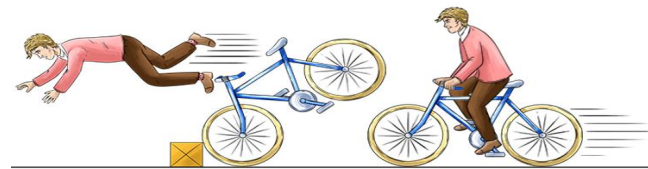
$$\sum \vec{F} = 0$$

- b) It is easier to work with the equation in terms of its component:

$$\sum \vec{F}_x = 0$$

$$\sum \vec{F}_y = 0$$

- ii. **Dynamic** equilibrium ($a = 0$)
→ body moving at a **uniform (constant) velocity**.



Newton's First Law of Motion



An object at rest
will remain at rest...

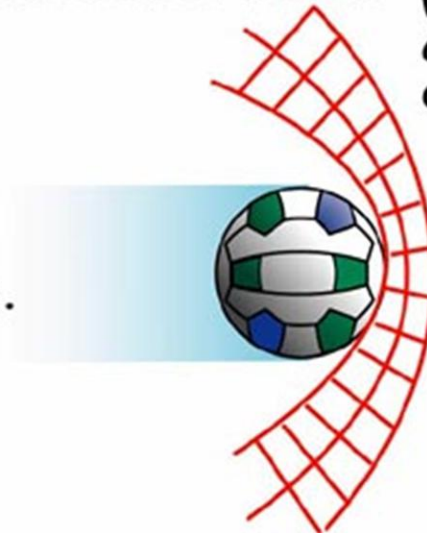


Unless acted on by
an unbalanced force.



An object in motion
will continue with
constant speed and
direction,...

... Unless acted on by
an unbalanced force.



NEWTON'S SECOND LAW

The **rate of change of momentum** of a moving body is **proportional to the net force** and is **in the same direction** as the net force action on it.

It can be represented by,

$$\sum \vec{F} \propto \frac{d\vec{p}}{dt}$$

Net force

The rate of change of momentum

a) Newton's second law can also be written as:

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

$$\sum \vec{F} = \frac{d(m\vec{v})}{dt}$$

$$\sum \vec{F} = \vec{v} \frac{dm}{dt} + m \frac{d\vec{v}}{dt}$$

Mass is constant,

$$\frac{dm}{dt} = 0$$

Therefore,

$$\sum \vec{F} = m \frac{d\vec{v}}{dt}$$

$$\sum \vec{F} = m\vec{a}$$

where $\sum F$: net force

m : mass

a : acceleration

NEWTON'S SECOND LAW

States that

The **acceleration** of an objects is **directly proportional** to the magnitude in the same direction as the net force and **inversely proportional** to the mass of the object

It can be expressed in equation as follows

$$a = \frac{F_{net}}{m}$$

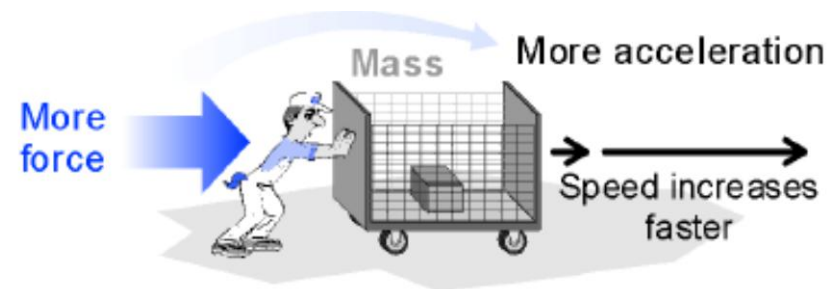
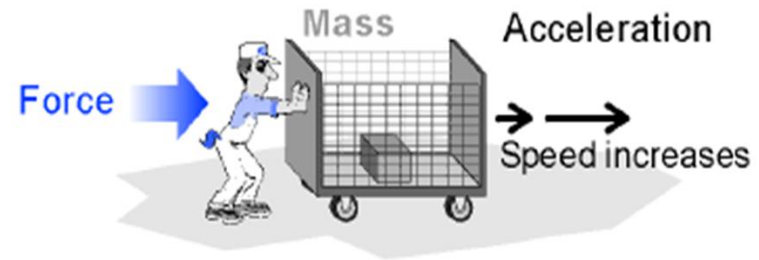
Acceleration (ms^{-2})

Net force (N)

mass (kg)

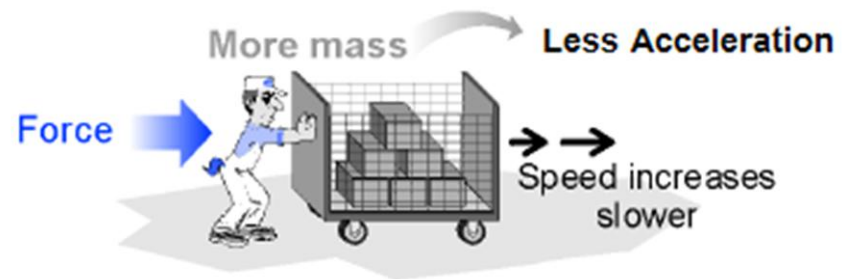
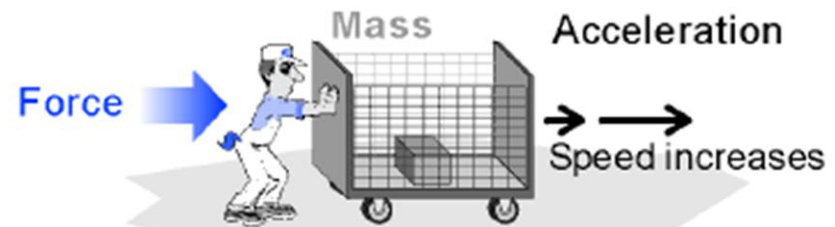
NEWTON'S SECOND LAW

If you apply more force to an object, it accelerates at a higher rate



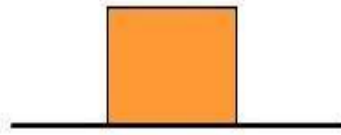
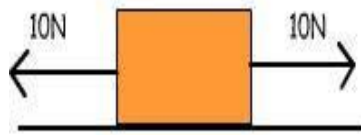
NEWTON'S SECOND LAW

If an object has more mass it accelerates at a lower rate because mass has inertia

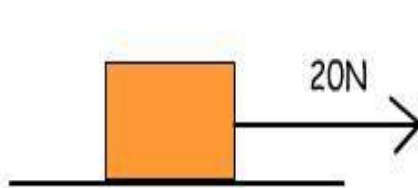
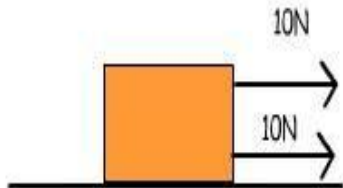


**Tips to write
Newton's
second law
equation**

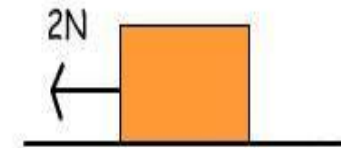
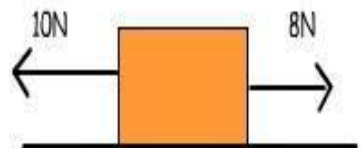
Identify the direction of $\blacklozenge$
 $\{ \text{Force(s) IN direction of } \blacklozenge \} - \{ \text{Force(s) OPPOSITE to } \blacklozenge \} = \blacklozenge \blacklozenge$



$$F_{\text{net}}=0$$

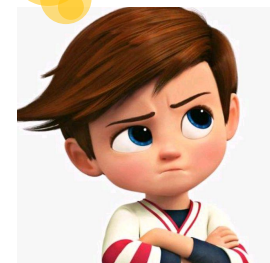


$$F_{\text{net}}=20\text{N}$$



$$F_{\text{net}}=2\text{N}$$

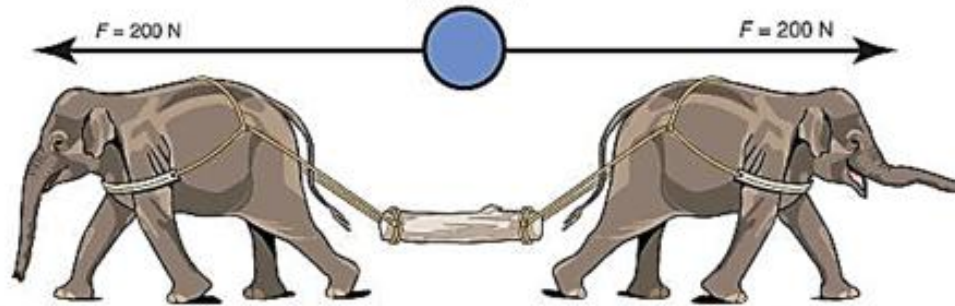
One newton (1 N) is defined as the amount of net force that gives an acceleration of 1 ms^{-2} to a body of 1 kg



Net force or resultant force ($\vec{F}_{net}$ @ $\sum \vec{F}$) is combination of all the forces acting on an object.

Newton's laws of motion

net force = 0



Forces in the Same Direction



Forces in Opposite Directions

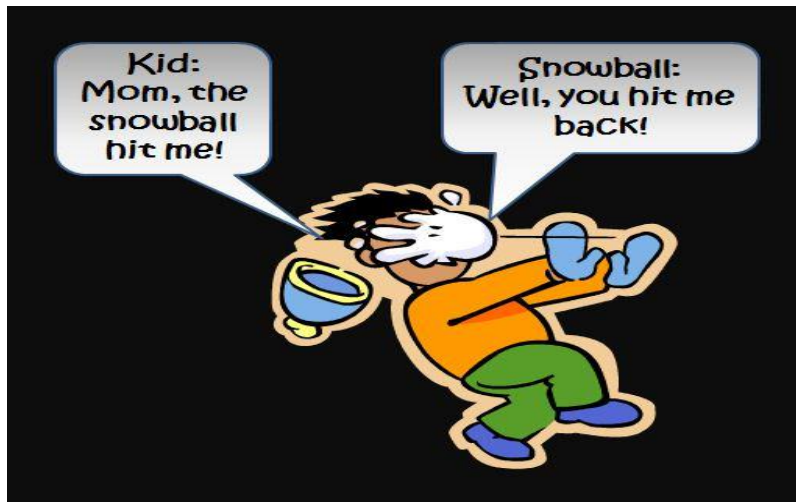


NEWTON'S THIRD LAW

State that

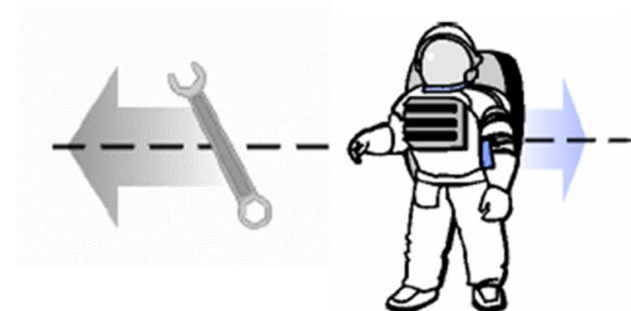
For **every action** there is a **reaction** that is **equal** in **magnitude** but in **opposite direction**

Newton's third law discusses pairs of objects and the interactions between them

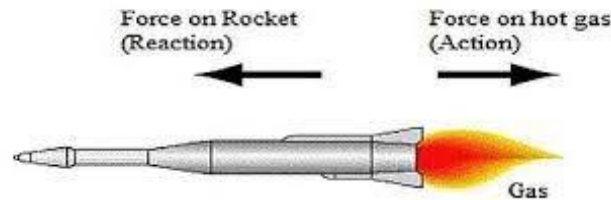
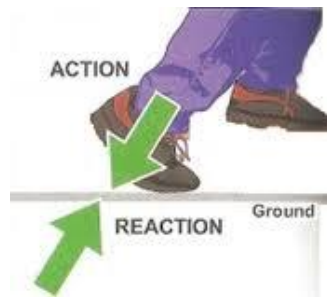


DID YOU KNOW THAT.....

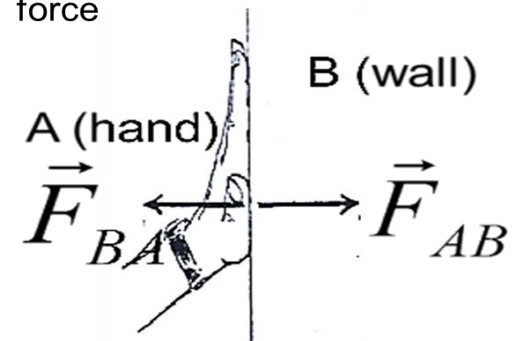
- The astronauts working on the space station have a serious problem when they need to move around in space. There is nothing to push on.
- The solution is by throw something opposite the direction you want to move
- This works because all forces always come in pairs



NEWTON'S THIRD LAW



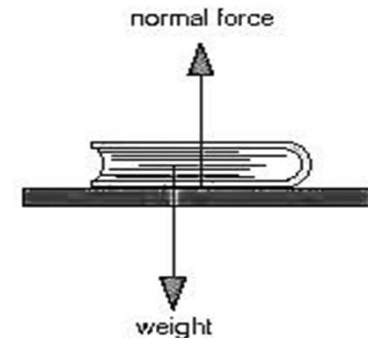
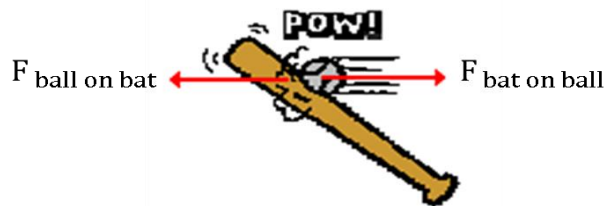
When you push on the wall, it will push back with the same force



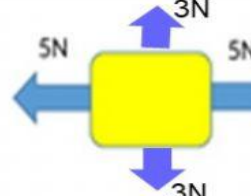
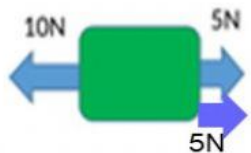
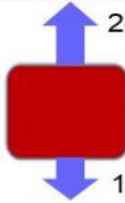
$\vec{F}_{BA}$ is also called normal force



When two bodies interact, the **action and reaction** forces **act on different bodies**. They **do not cancel out**.



CHECK YOUR UNDERSTANDING

	$F_{\text{net}} =$ N Direction :		$F_{\text{net}} =$ N Direction :
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Have you
get the
answer?



PROBLEM SOLVING STRATEGY TO APPLY NEWTON'S LAWS

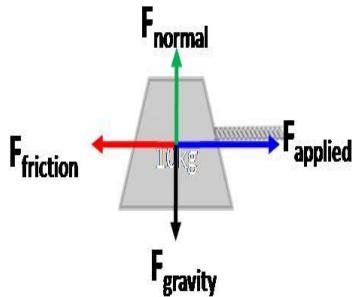
Identify all forces acting on the object

- Weight- have mass?
- Normal- touch with the surface?
- Tension- be tied by rope or cord?
- Friction- in the rough surface/ given μ

* Follow the direction for external force

Draw Free Body Diagram

- Using knot to represent the object and draw arrow and state symbol of forces



Resolve the forces into x and y direction

Net Force Problems

$$\sum F_x = F - f$$

$$\sum F_x = m a_x$$

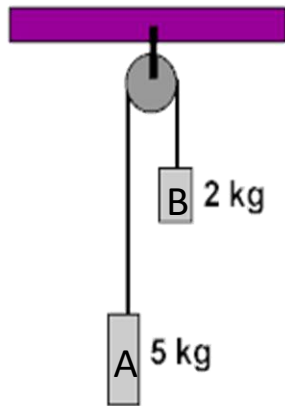
$$\sum F_x = F_x - f$$

$$F_x = F \cos \theta$$

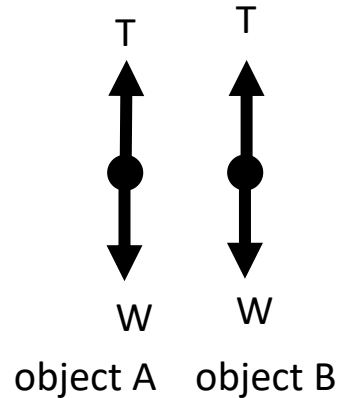
$$F_y = F \sin \theta$$

- Apply the suitable Newton's Law on both axes. Choose the motion as one of the axes

PROBLEM SOLVING STRATEGY TO APPLY NEWTON'S LAWS



From the situation given,
object A is more heavier to object B,



Apply Newton's Second Law

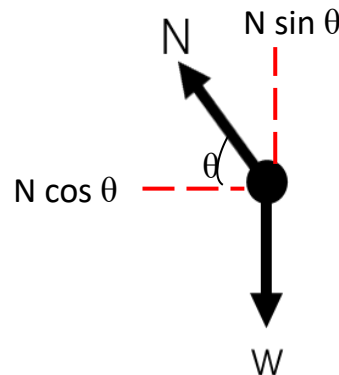
Object A: direction **downward**

$$F_{\text{net}} = ma$$
$$W - T = ma$$

Object B: direction **upward**

$$F_{\text{net}} = ma$$
$$T - W = ma$$

PROBLEM SOLVING STRATEGY TO APPLY NEWTON'S LAWS



From the situation, you see the man **move to the left**. He is dragging the luggage. Identify all forces acting on the luggage

Apply Newton's Second Law

x-axis: **direction to the left**

$$F_{\text{net}} = ma$$

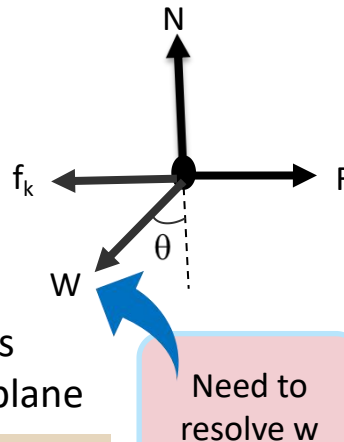
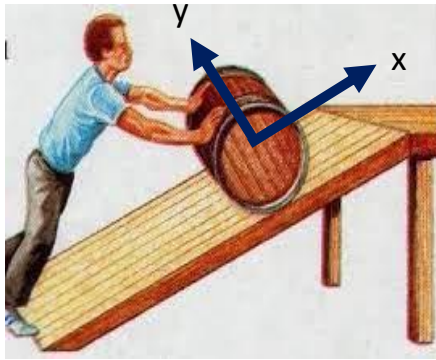
$$N \cos \theta = ma$$

y-axis: **no motion** in this axis

$$F_{\text{net}} = 0$$

$$N \sin \theta - w = 0$$

PROBLEM SOLVING STRATEGY TO APPLY NEWTON'S LAWS



From the situation, you see a man is **pushing** a drum on a **rough incline** plane

⚙️ Tips to draw FBD at incline plane

Normal force: must **PERPENDICULAR** with incline plane

Weight: directed **DOWNWARD** with the object

Apply Newton's Second Law

x-axis: object is push along the plane

$$F_{\text{net}} = ma$$

$$F - f_k - N \sin \theta = ma$$

Tips: start your equation by following direction of motion given in the question

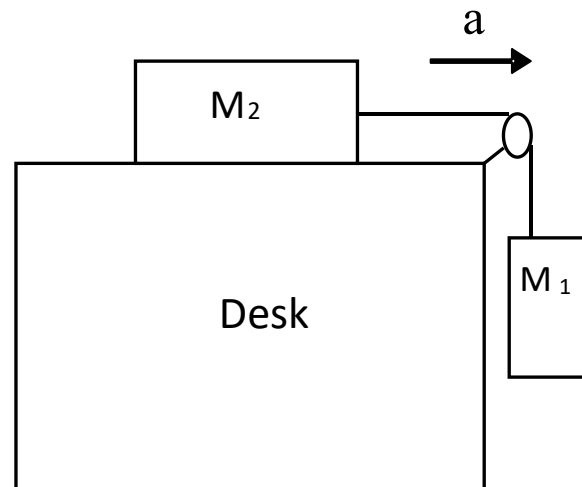
y-axis: **no motion** in this axis

$$F_{\text{net}} = 0$$

$$N - W \cos \theta = 0$$

CHECK YOUR UNDERSTANDING

Two blocks with masses $m_1 = 5 \text{ kg}$ and $m_2 = 1 \text{ kg}$ are connected by a string that passes over a pulley as shown in figure below. The coefficient of friction between the desk and m_2 is 0.01. Calculate the acceleration of the system.



Ans: $a = 8.159 \text{ m s}^{-2}$

CHECK YOUR UNDERSTANDING

Two wooden blocks, A and B of mass 2 kg and 4 kg respectively are connected by a rope. The system is placed on a rough surface and the friction forces on A and B are 2 N and 4 N. Find the acceleration of the system and the tension in the rope if the system is pulled by a force of 12 N.

Ans: $a = 1.0 \text{ m s}^{-2}$, $T = 4 \text{ N}$

Psst....
Something
AWESOME
Is coming!!

STAY TUNED!

